LegalRAG: Improved Techniques for Trademark Infringement Analysis with Triplet Loss Optimization

Ryan Crowley

Abstract

Trademark infringement poses a growing challenge for firms seeking to protect their brands. While recent developments in artificial neural networks (ANNs) have enabled sophisticated image-based monitoring and detection systems, these solutions neglect a fundamental aspect of the law: its nature as a "textual enterprise." This paper proposes a novel framework for analyzing trademark distinctiveness based on the *Abercrombie* factors—a critical threshold test for infringement cases. This approach combines retrieval augmented generation (RAG) with a triplet network of SentenceBERT encoders fine-tuned through triplet loss optimization.

1 Introduction

A firm's brand is often its most valuable asset (Keller and Lehmann, 2006). Unauthorized use of trademarks can erode brand equity, causing revenue decline[1] (Ertekin et al., 2018). The expanding access to online commerce and digital copying increases the burden on firms to monitor markets for these violations. Recent developments in ANNs (artificial neural networks), however, have provided trademark owners with novel monitoring solutions (Lin and Sam., 2020).

ANNs are computational processing systems that rely on layers of interconnected nodes to optimize specific tasks from input data representations (O'Shea and Nash, 2015). Training the network happens in two phases: the forward pass and back-propagation. In the forward pass, an activation function is applied to the weighted sum of each input node to determine its output (Nair and Hinton, 2010). See Figure 1 for an illustration.

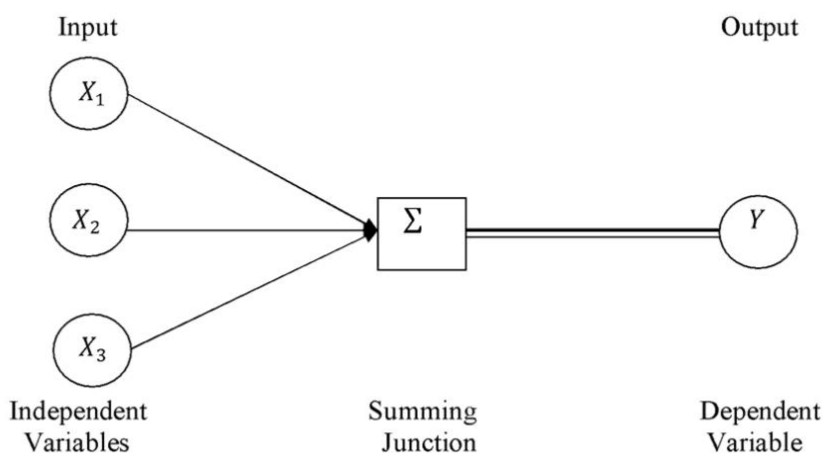


Figure 1. An illustration of a simple artificial neural network (Goodwin, 2014).

In supervised learning, the output is compared to a labeled, ground truth target to compute an "error". Unsupervised learning, on the other hand, uses unlabeled data. The error is computed based on the task's objective, such as clustering similar data closer together (Bromley et al., 1993). During back-propagation, the error is propagated backwards through the model, where a gradient decent algorithm updates the weight of the model (Rumelhart et al., 1986). The type of learning can be dependent on task requirements or availability of data, and some techniques blur these lines.

[1] The Lanham Act a trademark as any word, name, or symbol that serves as a firm's source identifier in commerce. 15 U.S.C. § 1127.

For trademark monitoring, the model's specific task is trademark retrieval—that is, searching a large database to find trademarks that are similar to a user-provided query image (Khanday and Shabir, 2021). Modern image retriever systems typically employ one of three ANN architectures: CNNs (convolutional neural networks) (LeCun et al., 1989), VITs (vision transformers) (Dosovitskiy et al., 2021), or CLIP (Contrastive Language-Image Pre-training) (Radfield et al., 2021) models.

These models capture relationships from the pixels of input images, allowing the model to detect similar patterns when deployed in a trademark monitoring tool (Tursun et al., 2019). Figure 2 is an example of input images for training.

Figure 2. Images contained in the METU benchmark trademark retrieval dataset (Tursun et al., 2019).

For the trademark owner, however, monitoring and detection only solves one half of the puzzle. Once a possible infringer is identified, the trademark owner has a choice to make: sue the possible infringer or risk degrading the value of its brand. Each year, over 3,000 trademark infringement lawsuits are filed in U.S. district courts, costing between $375,000 and $2 million if it proceeds to trial (Ertekin et al., 2018). While existing trademark retrieval systems provide advanced detection, they largely neglect a fundamental aspect of the law: its nature as a "textual enterprise" (Livermore and Rockmore, 2019).

Trademark infringement is defined by the "likelihood of confusion" standard, where the court examines whether there is a substantial risk that consumers will be confused as to the source, identity, sponsorship, or origin of the defendants' goods or services.[2] The "likelihood of confusion" standard requires applying a complex, multi-factored test (Lim, 2022). The threshold inquiry is over the mark's "distinctiveness." A mark is only eligible for trademark protection if it is considered to be distinctive. Distinctiveness also plays a role in other factors, such as the "strength" of a mark (Beebe and Hemphill, 2017).

[2] The Lanham Act prohibits the use of a registered mark in a manner "likely to cause confusion," 15 U.S.C. § 1114(1)(a).

In *Abercrombie & Fitch Co. v. Hunting World, Inc.*[3], the court established a spectrum of distinctiveness that continues to guide trademark analysis today (Lim, 2022). This spectrum classifies marks as generic, descriptive, suggestive, arbitrary, or fanciful, with each category requiring nuanced textual analysis of the mark in relation to its associated goods or services. A mark's classification depends on the relationships between the mark's literal meaning, its use in commerce, and its relationship to the goods or services it identifies. Because existing trademark retrieval systems are built to learn patterns from pixels contained in *images*, they serve little value to the firm considering legal action. Figure 3 illustrates the learning process for these systems.

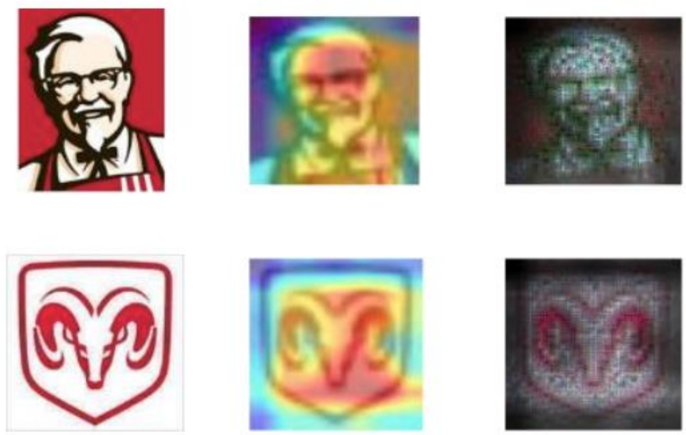


Figure 3. An illustration of the learning process for CNNs, VITs, and CLIP models.

This paper proposes a novel framework for threshold trademark infringement analysis based on the *Abercrombie* factors: a Retrieval-Augmented Generation (RAG) system utilizing a triplet network of Sentence-BERT (S-BERT) neural networks finetuned with triplet loss optimization. The rest of the paper is organized in the following. § 2 discusses the related works. § 3 presents the architecture and details. Limitations and policy are discussed in § 4. The conclusion is presented in § 5.

2 Related Works

2.1. Embeddings

Word embeddings are vector representations of words that encode semantic meaning. Performing mathematical operations like cosign similarity, Euclidian distance measurements, or vector algebra on these representations allows us to find patterns in the data (Choi, 2022). Early embedding efforts used one-hot encoding to represent each word as a feature vector. With this

[3] *Abercrombie & Fitch Co. v. Hunting World, Inc.,* 537 F.2d 4 (2d Cir. 1976).

method, every word addition dramatically increases the volume of the vector space. As data becomes sparse, computations become inefficient and ineffective, leading to the "curse of dimensionality" (Devlin et al., 2018).

A key innovation from Word2Vec (Mikolov et al., 2013) and GloVe (Pennington et al., 2014) was to use fixed-dimension dense embedding to reduce the complexity of the vector space. Word2Vec proposed an ANN with a skip-gram and continuous bag of words (CBOW) to utilize the network's hidden layer as a "proxy" for input representation. Because the CBOW ignores word order, the model struggled to capture global context (Devlin et al., 2018). Conversely, GloVe (Pennington et al., 2014) used global co-occurrence training to encode inputs as probabilities of co-occurrence. But these models are "static," meaning each input word has a fixed vector, regardless of context. This caused polysemy and homonymy issues—that is, the models couldn't distinguish the context of words that have the same spelling but have a similar or unrelated meaning (Devlin et al., 2018).

Because each word has a fixed vector regardless of context, static word embeddings have a limited ability to perform complex natural reasoning tasks. Nonetheless, they played a crucial role in deep learning development. and remain essential for downstream tasks, including machine learning, semantic analysis, ranking, classification, and retrieval augmented generation (Lewis et al., 2020); (Lee et al., 2024).

These static models have also enabled novel legal applications. Choi (2022) demonstrated GloVe's utility as a statutory interpretation tool, particularly for legal classification problems where 'Defendant's liability depends on whether X is a Y.' Livermore and Chau (2023) go into great depth on the history of the transition from "Law as Code" to "Law as Data." These works have inspired the proposed system in § 3.

2.2 Context Embedding

Transformers, introduced by Vaswani et al. (2017) for translation tasks, use a self-attention mechanism to process sequential data by weighing input tokens. While Vaswani et al. proposed an encoder-decoder architecture, the model family has split into two branches: GPT 3 (Brown et al., 2020), is a decoder-only model that autoregressively pre-trains to predict the next token in a sentence, excelling at text generation tasks; BERT (Devlin et al., 2018), an encoder-only model, bidirectionally pre-trains to predict masked random tokens, and is the dominant approach for embedding. SentenceBERT (Reimers & Gurevych, 2019) extends BERT by learning sentence-level embedding rather than token (word) level embedding.

Although embeddings are used initially for generation, all tasks involving text and language can be fundamentally categorized as either generative or embedding problems (Muennighoff et al., 2024). While most information on state-of-the-art commercial models remains propriety, recent papers (Lee et al., 2024) suggest an evolution towards "Generalist Embedding Models" that combine embedding tasks and generations tasks. This is accomplished with new architecture, different loss function, and synthetic training data.

2.3 Retrieval-Augmented Generation

Proposed by Lewis et al., (2020), retrieval-augmented generation (RAG) combines information retrieval with language generation to enhance outputs with external knowledge. Traditional language models are limited by their training data, which can lead to "hallucinations" when asked about information outside of their knowledge base. RAG addresses this by retrieving relevant passages during "inference" (when the model is running outside of training) from an external knowledge vector base. Lewis et al., demonstrated that this approach is particularly effective for domain-specific applications.

2.4 The Abercrombie Task

Guha et al. (2023) proposed LEGALBENCH as a legal reasoning benchmark consisting of 162 tasks covering six different legal reasoning types. The Abercrombie task, based on *Abercrombie & Fitch Co. v. Hunting World, Inc*., requires LLMs to classify product-trademark pairs into five distinctiveness categories that determine trademark protection eligibility:

1. **Generic.** The name connotes the basic nature of the product/service or merely refers to the class of goods. Other example? The mark "Salt" for packaged sodium chloride would be generic under *Abercrombie* because "salt" is the common name for sodium chloride.
2. **Descriptive.** The name identifies a characteristic or quality of an article or service, such as color, odor, function, dimensions, or ingredients.
3. **Suggestive.** The name suggests, rather than describes some particular characteristic of the goods or services to which it applies. An essential aspect of suggestive names is that it requires the consumer to exercise the imagination in order to draw a conclusion as to the nature of the goods and services.
4. **Arbitrary.** The name is a "real" word but seemingly "arbitrary" with respect to the product or service. For example, the mark "Apple" for a software company is arbitrary, because apples are unrelated to software.
5. **Fanciful.** A name is fanciful if it is entirely made up, and not found in the English dictionary.

The LLM must explain the classification and is evaluated on: (1) whether the explanation is correct, and (2) whether it contains the necessary analysis to link facts to law (Guha et al., 2023). The proposed method is designed around optimizing this task.

3 Method

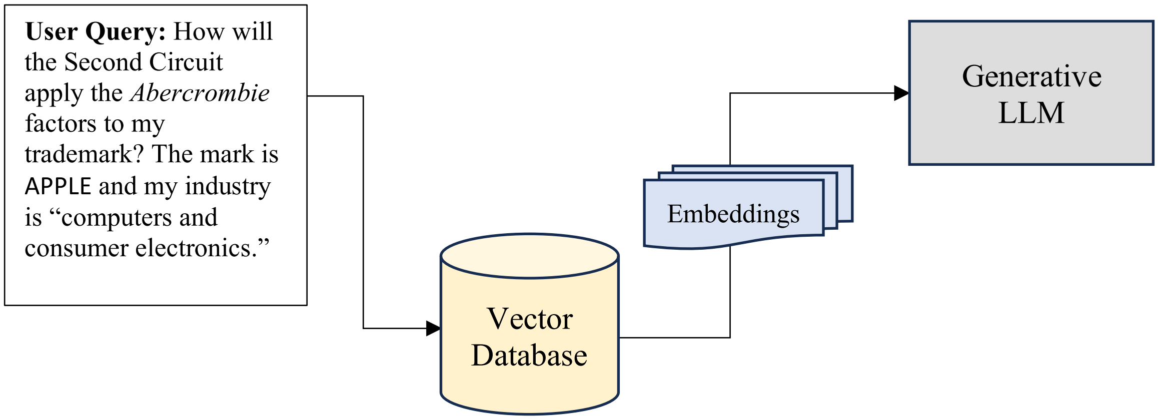


Figure 4. The proposed methodology includes a RAG pipeline with a fine-tuned embedding model (**Figure 5**).

3.1 Retrieval-Augmented Generation

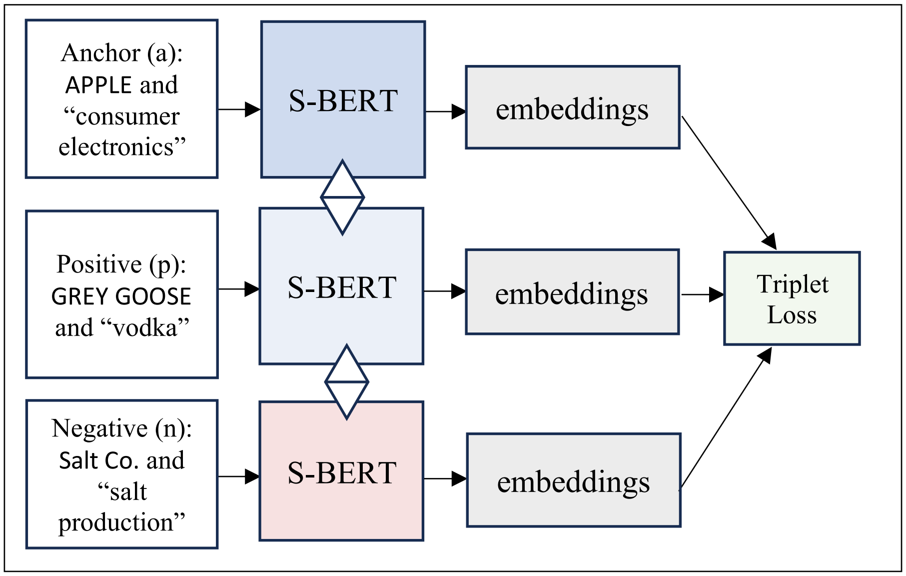
A fundamental issue in trademark infringement litigation is that courts apply the likelihood of confusion factors differently. Additionally, while distinctiveness is considered a threshold issue in all circuits, the role of distinctiveness in other factors of the likelihood of confusion analysis varies significantly by circuit (Beebe and Hemphill, 2017).[4] Thus, the key consideration for the trademark owner's litigation strategy is where to file.

The RAG component is the "organizing principle" for this system (Livermore et al., 2021). The component operates in two distinct stages. During the retrieval stage, the system searches the vector database of federal court decisions and USPTO Trademark Trial and Appeal Board (TTAB) opinions applying the *Abercrombie* factors. The retriever employs dense passage retrieval (DPR) to identify relevant legal precedents from the queried circuit. (Karpukhin et al., 2020). Unlike traditional keyword search, DPR captures semantic relationships between the legal concepts in the input query, enabling the identification of relevant precedents, even when they use different terminology.

When analyzing whether a product-trademark pair like APPLE for the "computers and consumer electronics industry", DPR allows the system to retrieve cases discussing similar arbitrary marks in unrelated fields. At the generation stage, the language model uses the retrieved embeddings to inform the analysis of the Abercrombie factors, ensuring that the system's reasoning aligns with the established Circuit's principles. See Figure 4 for an illustration.

[4] Beebe B, Hemphill CS. The scope of strong marks: should trademark law protect the strong more than the weak? NY Univ Law Rev. 2017;92: at 1345.

3.2 SentenceBERT Vector Embedding



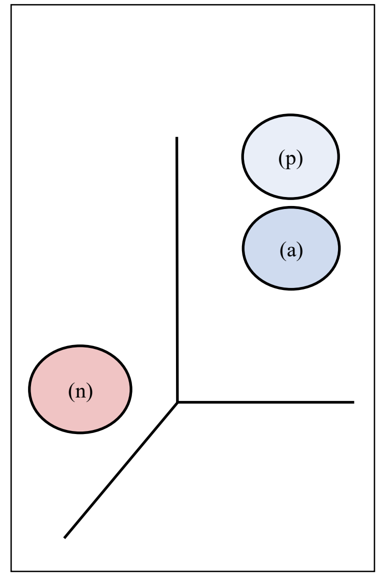


Figure 5. A triplet network of SentenceBERT encoders.

Figure 6. Embeddings.

A self-supervised triplet network of fine-tuned SentenceBERT encoders provide the embeddings for the vector database. Because SentenceBERT combines the contextual understanding of BERT with the concise, meaningful sentence representations, it is preferred for a RAG implementation (Kumawat, 2024). SentenceBERT maps user queries and legal text into high-dimensional vectors (768 dimensions). The three parallel triplet networks share identical weights, and the triplet loss function operates on the three inputs:

- Anchor (a): The product-trademark pair under analysis.
- Positive (p): A similar product-trademark from the same *Abercrombie* category
- Negative (n): A trademark from a different *Abercrombie* category

Following Reimers and Iryna Gurevych., (2019), the triplet loss tunes the network such that the distance between (a) and (p) is smaller than the distance between (a) and (n). The following loss function is minimized:

$$L = \max(\|f(a) - f(p)\|^2 - \|f(a) - f(n)\|^2 + \text{margin}, 0)]$$

- f(x) represents the embedding function
- $\|\cdot\|^2$ denotes Cosine distance
- margin refers to bias (typically set to 1.0)

Triplet loss is valuable for classifying the product-trademark pair into the *Abercrombie* factors because it creates a semantic space where trademarks within the same category cluster together while maintaining clear separation between different categories. This mirrors reality where distinctiveness categories form a spectrum but require clear boundaries for practical application (Lim, 2022).

This approach also excels at building an embedding space that can handle the nuanced differences between. For instance, a "fanciful" mark like EXXON will be considered inherently distinctive because it doesn't exist in the English language. Distinctiveness can play a role in the "strength factor," which is another part of the likelihood of confusion test. The First and Tenth Circuits, however, only consider strength as the least important factor (Beebe and Hemphill, 2017).[5]

3.3 Training

To model is trained using a combination of supervised and self-supervised learning techniques. Initially, the model uses labeled data from USPTO classifications and court decisions that act as a ground truth of *Abercrombie* classifications. The training process, however, introduces a self-supervised element through "triplet mining." While the overall categories and relationships are "supervised," the model participates in selecting which labels to learn during mining (Shah, 2023). To build an effective embedding space, it is optimal for the model to mine as many "hard" cases as possible. Hard cases are those where the anchor, positive, and negative samples would be all relatively close on the distinctiveness spectrum.

4 Limitations

4.1 Data Limitations

The effectiveness of the proposed framework is fundamentally constrained by data challenges. Like all machine learning systems, the model inherits biases present in its training data. This manifests in several ways:

The S-BERT models used were pre-trained on massive amounts of internet data (Devlin et al., 2018). These models have been known to perpetuate discriminatory biases (May et al., 2019). Here, even though we are "fine-tuning" the model with legal corpus, the original pre-training data may encode societal biases about brands, products, and commerce that could influence the *Abercrombie* analysis. For instance, pre-encoded assumptions about the breadth of a particular firm's commerce might lead to a faulty prediction when that firm uses the system.

There is likely a limited availability of high-quality training data in existence. From the roughly 3000 trademark cases that are filed per year, the chances that (1) it goes to trial, and (2) the court

---

[5] Beebe B, Hemphill CS. The scope of strong marks: should trademark law protect the strong more than the weak? NY Univ Law Rev. 2017;92: at 1349, FN 40.

writes about it in an opinion are quite slim. Lim (2022) asserts that in published opinions, the courts expressly discuss “distinctiveness” in 70% of cases. Thus, there is likely only a couple of hundred product-trademark training samples available for the entire federal circuit. There is probably fewer “good” samples for triplet mining. An alternative would be to have a human evaluator label anchor-positive-negative triplets from the USPTO dataset, but this obviously increases the chances of introducing bias.

Furthermore, novel trademark disputes involving new technology, new business models, or new language may fall outside the model's learned embeddings. A term that was once “fanciful” (doesn’t exist in the human language) might become “generic” (no trademark protection) over time— “Kleenex” and “thermos,” for example. Assuming the model worked, it would likely be able to identify similar historical cases. But it may struggle to adapt its analysis to “known unknowns” and “unknown unknowns.”

The subjective nature of trademark law itself may result in the often quoted “garbage in, garbage out” analogy. First, there is potential for significant noise in the training data. As noted above, the role distinctiveness plays in the broader likelihood of confusion analysis significantly varies between circuits and judges. What one judge considers "suggestive," another might classify as "descriptive." Lim (2022) notes that determining trademark distinctiveness is “so subjective that [it] poses challenges to the development of AI in trademark law.”[6] Second, if there is limited data on a particular judge, then the model would probably be of limited use.

A possible solution to the limitation of data would be synthetic data generation. Lee et al., (2024) proposes a method for mining pairs using a generative LLM by creating adversarial examples from existing real pairs (here, the metric learning loss was contrastive, which only requires a positive and negative pair). This method could also be used to generate entire new cases for each circuit. But this still doesn’t address the bias or “garbage in, garbage out” problems, as the synthetic data would be created from the existing case law.

4.2 Practical Limitations

There are also practical limitations to this particular system. It would be expensive and time consuming to collect triplet samples, execute the training, and then deploy the model into a software tool. Would this be worth the cost? Likely not. As Guha et al. (2023) explain in LEGALBENCH, the *Abercrombie* analysis is an “easy task for lawyers with a basic training in intellectual property law. It is unlikely that LLMs will be called on to perform this task in the

[6] Lim J. Trademark confusion revealed: an empirical analysis. Am U Law Rev. 2022;71: at FN 469.

actual practice of law."[7] Thus, the costs required to build and implement such a system would probably render the product useless.

On the other hand, it may be morally and ethically justified to build such a tool. From a deontological perspective (Livermore et al., 2021), one could argue that a firm has a fundamental right to protecting its brand and use of trademarks. For those who wish to use the trademark, there is a moral duty to obtain explicit permission its owner. And, assuming this system could be built well, it could help bring about just enforcement by providing the trademark owner with critical litigation insights.

5. Conclusion

This paper has proposed a novel framework for analyzing trademark distinctiveness using a retriever-augmented generation pipeline that is powered by triplet networks of SentenceBERT encoders. Important considerations emerge from this analysis.

First, the success of such a system working heavily depends on data quality and quantity. The inherent subjectivity of trademark law crates challenges for training. Even if we could train an operational model, the variability of standards across different circuits and judges may render its application mute. Second, the practical limitations of this specific legal task suggest limited commercial value for trademark owners. The relative simplicity of applying the *Abercrombie* factors, combined with the likely costs to build, train, and implement such a tool, are likely not worth any perceived benefits. As the market for artificial intelligence consolidates around 2-3 companies, perhaps this could be extended to all tools.

Research for the proposed model, however, revealed important implications for "Law Search" (Livermore et al., 2021). In class, we discussed how the "curse of dimensionality" was particularly relevant for long, unstructured legal texts. But retriever-augmented generation seems to show significant potential for mitigating the curse (Lee et al., 2024). This area warrants further research.

Word Count: 4180

[7] Guha A, Gu Y, Kumar S, Zhao J, Zhao R, Wang A, et al. Legalbench: a collaboratively built benchmark for measuring legal reasoning in large language models. In: Proceedings of the Conference on Neural Information Processing Systems, Datasets and Benchmarks Track; 2023. At 45.

References

Legal:

Lim J. Trademark confusion revealed: an empirical analysis. Am U Law Rev. 2022;71:1285-1346. https://papers.ssrn.com/sol3/papers.cfm?abstract_id=4088929

Livermore MA, Rockmore DN. Law as data: computation, text, and the future of legal analysis. Santa Fe (NM): Santa Fe Institute Press; 2019.

Choi J. Measuring clarity in legal text. U Chi Law Rev. Forthcoming 2024;91.

Livermore MA, Chau N. Studying judicial behavior with text analysis. In: Oxford handbook of comparative judicial behavior. Oxford (UK): Oxford University Press. Forthcoming 2024.

Beebe B, Hemphill CS. The scope of strong marks: should trademark law protect the strong more than the weak? NY Univ Law Rev. 2017;92:1339-1404. https://papers.ssrn.com/sol3/papers.cfm?abstract_id=3073130

Livermore MA, Rockmore DN, Hamann H, Osenga K. Law search in the age of the algorithm. Mich State Law Rev. 2021;2020:1183-1220.

Guha A, Gu Y, Kumar S, Zhao J, Zhao R, Wang A, et al. Legalbench: a collaboratively built benchmark for measuring legal reasoning in large language models. In: Proceedings of the Conference on Neural Information Processing Systems, Datasets and Benchmarks Track; 2023. https://arxiv.org/abs/2308.11462

Computer Science:

O'Shea K, Nash R. An introduction to convolutional neural networks. ArXiv preprint arXiv:1512.03385. 2015. https://arxiv.org/abs/1512.03385

Nair V, Hinton GE. Rectified linear units improve restricted Boltzmann machines. In: Proceedings of the 27th International Conference on Machine Learning; 2010. p. 807-814. https://doi.org/10.1109/ICML.2010.101

Bromley J, Guyon I, LeCun Y, Säckinger E, Shah R. Signature verification using a siamese time delay neural network. In: Advances in Neural Information Processing Systems 6; 1993. https://doi.org/10.1109/IJCNN.1993.3878947

Rumelhart DE, Hinton GE, Williams RJ. Learning representations by back-propagating errors. Nature. 1986;323:533-536. https://doi.org/10.1038/323533a0

Khanday A, Shabir A. [Title unknown]. [Journal unknown]. 2021.

LeCun Y, Boser B, Denker JS, Henderson D, Howard RE, Hubbard W, et al. Backpropagation applied to handwritten zip code recognition. Neural Comput. 1989;1(4):541-551. https://doi.org/10.1162/neco.1989.1.4.541

Dosovitskiy A, Beyer L, Kolesnikov A, Weissenborn D, Zhai X, Unterthiner T, et al. An image is worth 16x16 words: transformers for image recognition at scale. In: International Conference on Learning Representations; 2021. https://arxiv.org/abs/2010.11929

Radford A, Kim JW, Hallacy C, Ramesh A, Goh G, Agarwal S, et al. Learning transferable visual models from natural language supervision. In: Proceedings of the 38th International Conference on Machine Learning; 2021. https://arxiv.org/abs/2103.00020

Tursun O, Deniz E, Yucel S. 2019.

Devlin J, Chang MW, Lee K, Toutanova K. BERT: pre-training of deep bidirectional transformers for language understanding. ArXiv preprint arXiv:1810.04805. 2018. https://arxiv.org/abs/1810.04805

Mikolov T, Sutskever I, Chen K, Corrado GS, Dean J. Distributed representations of words and phrases and their compositionality. In: Advances in Neural Information Processing Systems 26; 2013. https://arxiv.org/abs/1310.4546

Pennington J, Socher R, Manning CD. GloVe: global vectors for word representation. In: Proceedings of the 2014 Conference on Empirical Methods in Natural Language Processing; 2014. https://arxiv.org/abs/1408.5882

Lee K, Gunasekaran A, Park S. [Title unknown]. [Journal unknown]. 2024.

Vaswani A, Shazeer N, Parmar N, Uszkoreit J, Jones L, Gomez AN, et al. Attention is all you need. In: Advances in Neural Information Processing Systems 30; 2017. https://arxiv.org/abs/1706.03762

Lewis M, Liu Y, Goyal N, Ghazvininejad M, Mohamed A, Levy O, et al. BART: denoising sequence-to-sequence pre-training for natural language generation, translation, and comprehension. In: Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics; 2020. https://arxiv.org/abs/1910.13461

Brown TB, Mann B, Ryder N, Subbiah M, Kaplan J, Dhariwal P, et al. Language models are few-shot learners. In: Advances in Neural Information Processing Systems 33; 2020. https://arxiv.org/abs/2005.14165

Reimers N, Gurevych I. Sentence-BERT: sentence embeddings using Siamese BERT-networks. In: Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing; 2019. https://arxiv.org/abs/1908.10084

Muennighoff N, Chen P, Patel M. 2024. https://arxiv.org/pdf/2402.09906

Karpukhin V, Oguz B, Min S, Lewis P, Wu L, Edunov S, et al. Dense passage retrieval for open-domain question answering. In: Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing; 2020. https://arxiv.org/abs/2004.04906

May C, Wang A, Bordia S, Bowman SR, Rudinger R. On measuring social biases in sentence encoders. In: Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies; 2019. p. 622-628. https://aclanthology.org/N19-1063/

Economics:

Keller KL, Lehmann DR. Brands and branding: research findings and future priorities. Marketing Sci. 2006;25(6):740-759. https://doi.org/10.1287/mksc.1050.0153

Ertekin S, Smith B, Rudin C. [Title unknown]. [Journal unknown]. 2018.

Web:

(Kumawat, 2024). https://medium.com/@tejpal.abhyuday/retrieval-augmented-generation-rag-from-basics-to-advanced-a2b068fd576c

(Shah, 2023). https://www.v7labs.com/blog/triplet-loss